TECH ARENA

ROUND 3 — CODING PROBLEM PAPER

Easy • Moderate • Hard

Instructions

• Write a complete program for each problem in the programming language announced by the organizers.
• Read input from standard input and print only the required output.
• Test your program against the sample case and consider edge cases.
• Solutions are intentionally not included in this participant copy.

Participant name: _____ College: _____

EASY

Focus: loops, conditions, arrays, and strings.

1. The Missing Number

Write a program that reads N and N−1 distinct integers from the range 1 to N. Print the missing number.

Sample Input

```
Input:
5
1 2 4 5
```

Sample Output

```
Output:
3
```

Constraints: 2 ≤ N ≤ 100,000

2. Count the Vowels

Write a program that counts the vowels (A, E, I, O, U) in a given line of text. Uppercase and lowercase vowels both count.

Sample Input

```
Input:
Tech Arena
```

Sample Output

```
Output:
4
```

Constraints: Text length: 1 to 10,000 characters

3. Second Largest Distinct Number

Write a program to print the second-largest distinct value in an array. If no second-largest distinct value exists, print -1.

Sample Input

```
Input:
6
4 9 9 2 7 7
```

Sample Output

```
Output:
7
```

Constraints: 1 ≤ N ≤ 100,000

MODERATE

Focus: stacks, frequency counting, and array traversal.

4. Balanced Brackets

Write a program that checks whether a string containing only (), [] and {} has correctly matched and ordered brackets. Print YES if balanced; otherwise print NO.

Sample Input

```
Input:
{[()]}
```

Sample Output

```
Output:
YES
```

Constraints: String length: 1 to 100,000

5. Longest Increasing Segment

Given an array, print the length of the longest contiguous segment in which every number is strictly greater than the previous number.

Sample Input

```
Input:
7
1 2 2 3 4 5 1
```

Sample Output

```
Output:
4
```

Constraints: $1 \leq N \leq 200,000$

6. First Non-Repeating Character

Print the first character in a string that occurs exactly once. Treat uppercase and lowercase as different characters. If none exists, print -1.

Sample Input

```
Input:
swiss
```

Sample Output

```
Output:
w
```

Constraints: String length: 1 to 100,000

HARD

Focus: dynamic programming, graph traversal, and efficient algorithms.

7. Minimum Coins

Given coin denominations and a target amount, find the minimum number of coins needed. You may use each denomination unlimited times. Print -1 if the amount cannot be formed.

Sample Input

```
Input:
3 11
1 5 7
```

Sample Output

```
Output:
3
```

Constraints: 1 ≤ number of denominations ≤ 100; target amount ≤ 100,000

8. Shortest Path in a Grid

A grid uses 0 for open cells and 1 for blocked cells. Starting at the top-left cell, find the minimum number of up/down/left/right moves to reach the bottom-right cell. Print -1 if no path exists or an endpoint is blocked. Count moves, not cells.

Sample Input

```
Input:
3 3
0 0 1
1 0 0
0 0 0
```

Sample Output

```
Output:
4
```

Constraints: 1 ≤ rows, columns ≤ 500

9. Maximum Sum Subarray

Given an array containing positive, negative or zero integers, find the maximum sum of any non-empty contiguous subarray.

Sample Input

```
Input:
8
-2 -3 4 -1 -2 1 5 -3
```

Sample Output

```
Output:
7
```

Constraints: 1 ≤ N ≤ 200,000; each value is between -10^9 and 10^9

Suggested judging criteria

- Correct output on sample and hidden test cases
- Handling of edge cases
- Time and memory efficiency
- Clear, readable code

Organizer note: Add hidden test cases before the competition. Make the permitted languages and time limit clear to all participants.